

Machine Learning Project Report

Waste Type Identification

Group: 2026_MLinf_gr41 — Gioia Errico, Cutolo Raffaele

1 Dataset Motivation and Split Strategy

The dataset presents two main problems: a strong class imbalance (the *Clothing* class alone accounts for 47.1% of the images, while *Undifferentiated* is only 4.5% — a $10.5\times$ ratio) and high variability in image conditions. A purely random train/validation split would risk under-representing the smaller sub-classes (e.g. glass/green, papery/cardboard) in validation, making the validation metric unreliable exactly where the model is weakest.

We therefore built a custom stratified 80/20 split that preserves the proportion of every macro-class and every sub-class in both parts (e.g. Clothing: 5842 training images / 1460 validation images, with the same 20% ratio also applied to its “clothes” and “shoes” sub-classes). In this way our validation Balanced Accuracy — the official metric — is a reliable indicator for all 8 classes. The split is saved as `split.csv` and reused, unchanged, in every experiment in this report.

2 Baseline Model

Configuration. We fine-tuned a ResNet18 pre-trained on ImageNet, replacing the last layer with an 8-output linear head. All images are resized to 224×224 without cropping: our exploratory analysis showed highly variable aspect ratios, and cropping would often have cut away part of the object. No data augmentation at this stage. Training: SGD (momentum 0.9, weight decay $1e-4$, learning rate 0.01), cosine-annealing scheduler, 15 epochs, plain Cross-Entropy, keeping the checkpoint from the epoch with the best validation.

Results and error analysis. Validation Balanced Accuracy: **0.967**, with convergence within ~ 10 epochs followed by a plateau (a first sign of overfitting). The weakest class is *Plastic* (TPR 0.89), confused mainly with *Glass* — a plausible error, since transparent plastic and glass containers look visually similar. Clothing, Organic and Papery are nearly perfect. These observations guide the subsequent experiments.

3 The Generalization Gap: an Out-of-Distribution Probe

The 0.967 value must be read with caution. Our validation images come from the same cameras, backgrounds and post-processing as the training ones, so this number measures how well the model fits *this specific dataset* — not how well it recognizes waste in general. The assignment, on the other hand, specifies that the private test set is acquired with different cameras, under very different conditions, and possibly post-processed. We keep three numbers distinct throughout the report: the *official metric* (8-class Balanced Accuracy on the private test set, unknown to us), our *validation score* (its in-distribution proxy) and *RealWaste* (a 7-class out-of-distribution diagnostic, introduced below).

To check whether the baseline had learned dataset-specific shortcuts, we evaluated it — without any re-training — on **RealWaste**, an independent public dataset of waste images. RealWaste was used exclusively for evaluation and was never added to the training data, in full compliance with the rule forbidding any extension of the training set. It has no Battery class, so the check covers the seven shared classes, with its labels mapped onto our taxonomy.

Result: Balanced Accuracy collapses from 0.967 to **0.455**. The drop is highly uneven: Clothing stays high, Organic and Plastic fall sharply, and Undifferentiated becomes almost random (TPR 0.04) despite being one of the best classes in-distribution (0.976). The confusion matrix on RealWaste shows a systematic pattern: the Clothing column acts as an *attractor* — errors from nearly every class end up there. We identify two concurrent causes: (i) Clothing accounts for $\sim 47\%$ of the training set, so an unweighted loss gives the model a strong prior towards it, and when the model is uncertain it falls back on the most frequent class; (ii) Clothing is visually heterogeneous (garments and shoes of any color and shape), so it occupies a large region of feature space and works as a catch-all container.

Stability check (5-fold cross-validation). To verify that the validation score is not the result of luck on a single split, we ran a stratified 5-fold CV with exactly the same recipe: 0.960 ± 0.005 (fold 1 coincides with our split and reproduces 0.965), with every per-class TPR stable (std dev < 0.022). The in-distribution estimate is therefore reliable — and consequently the OOD collapse cannot be noise: it is a real effect of the distribution shift. *This gap is the central result of the project*: the goal of everything that follows is to close it, not to further inflate an already saturated validation score. We use RealWaste as a deliberately pessimistic diagnostic lens, alongside — not instead of — the in-distribution metric. A caveat on its absolute value: two label mappings are noisy (Textile Trash→Clothing, Miscellaneous→Undifferentiated), so the RealWaste number should be used mainly to compare models rather than as absolute accuracy; since the mapping is identical for all runs, the comparisons remain fair nonetheless.

4 Experimental Plan

We vary one factor at a time. First we select the *training recipe* (data augmentation and imbalance handling) on a fixed ResNet18, because these countermeasures address properties of the dataset, not of a specific model. Only once the best recipe is fixed do we compare *backbones* under that frozen recipe, since a ranking obtained with a weaker recipe might not transfer. All recipe experiments share the same split, optimizer, schedule, epochs and seed; the only variation is the factor under examination, so any difference is attributable to that single factor.

5 Experiment 1: Photometric Data Augmentation

Hypothesis. If the baseline relies on low-level acquisition cues — color balance, sharpness, JPEG artifacts, sensor noise — specific to this dataset, then perturbing precisely these cues during training should force the model to ignore them, raising the OOD score even if validation stays unchanged.

Design. A single parameter, AUG_LEVEL, applied only to the training set: **light** = horizontal flip, small rotations, mild ColorJitter; **strong** = **light** + RandomGrayscale, GaussianBlur, simulated JPEG compression, stronger jitter, RandomErasing.

Model	Validation (in-distr.)	RealWaste (OOD)
baseline	0.967	0.455
light	0.968	0.459
strong	0.954	0.432

Table 1: Balanced Accuracy on the two sets. RealWaste is 7-class (without Battery).

Reading. The hypothesis is not confirmed. **light** changes nothing real: the OOD gain (+0.004) is below the CV noise of ± 0.005 . **strong** is genuinely worse on both sets (OOD -0.023 , about 4 times the noise). In-distribution, **strong** mainly hurts the small, hard classes (Plastic $0.91 \rightarrow 0.84$): grayscale together with heavy jitter and blur erases precisely the color and glossiness cues that distinguish plastic, glass and organic. On RealWaste the effect is a redistribution — some classes improve (glass $0.46 \rightarrow 0.66$), others collapse (plastic $0.35 \rightarrow 0.18$) — which is negative overall.

Conclusion. Augmentation intensity behaves like an inverted U: a little is harmless, too much destroys the signal. More importantly: photometric corruption simulates *how the photo was taken*, not *how the object looks*: the real shift towards RealWaste concerns the object’s appearance and context (clean product photos vs. real dirty waste on cluttered backgrounds). We keep **light** as a harmless default, hold the geometric family in reserve (Section 7), and deliberately postpone to Section 10 the central, not-yet-tested part of the intensity curve — a mild, calibrated dose.

6 Experiment 2: Class Imbalance Handling

Hypothesis. If the Clothing prior identified in Section 3 is a main cause of the gap, neutralizing it should raise the OOD TPR of the minority classes. In-distribution we expect no change by construction, because Balanced Accuracy already weights all classes equally.

Design. Baseline entirely frozen, deliberately *without* augmentation (so as not to mix with the effect

under examination). A single factor, the rebalancing mechanism: `weighted_loss` = Cross-Entropy with per-class weights inversely proportional to class frequency; `sampler` = WeightedRandomSampler that oversamples minority classes so that each batch is approximately balanced.

Class	baseline (OOD)	weighted_loss (OOD)	sampler (OOD)
clothing	0.899	0.874	0.893
glass	0.462	0.505	0.367
metal	0.511	0.546	0.518
organic	0.388	0.401	0.115
papery	0.534	0.483	0.662
plastic	0.349	0.296	0.296
undifferentiated	0.042	0.067	0.024
OOD Balanced Accuracy	0.455	0.453	0.411

Table 2: Per-class OOD TPR on RealWaste. In-distribution all variants are indistinguishable from the baseline (0.961–0.962 vs 0.960 ± 0.005).

Reading. `Weighted_loss` mechanically does what it was designed for — the Clothing attractor weakens and several minority classes gain — but the gains are cancelled by the losses on Plastic and Papery: net $0.453 \approx 0.455$. Re-weighting shifts the decision boundary, but it operates on features that are already wrong out-of-distribution: it changes *which wrong class* a sample falls into, not *whether its features are correct*. `Sampler` is clearly worse (0.411), and the reason is instructive: without augmentation, oversampling shows the model the same few minority images many times, amplifying the memorization of their acquisition shortcuts; Organic collapses ($0.388 \rightarrow 0.115$).

Conclusion. Rebalancing does not close the gap. A lesson to carry forward: together with Experiment 1, the diagnosis converges; the dominant factor is the shift in object appearance and context, which neither photometric augmentation nor rebalancing can touch.

7 Experiment 3: Geometric and Structural Augmentation

Hypothesis. Geometric transformations change *how the object is framed* — its position, scale and visible portion — rather than how the photo was taken. If the real shift concerns framing and context, this family (held in reserve since Experiment 1) should be the one that helps.

Design. The reference here is `light`, not the bare baseline, so the isolated factor is the geometric perturbation. `geo_rrc` = `light` + RandomResizedCrop; `geo_rrc_cutmix` = the same + CutMix, which pastes a part of one image onto another and mixes the labels proportionally. The validation transform stays unchanged, keeping all numbers comparable.

Configuration	Validation (in-distr.)	RealWaste (OOD)
baseline	0.967	0.455
light (reference)	0.968	0.459
geo_rrc (light + RRC)	0.964	0.426
geo_rrc + CutMix	0.965	0.331

Reading. Not confirmed — the sharpest negative result so far. Validation stays unchanged (~ 0.965 everywhere), while OOD never surpasses `light`: RRC lowers it slightly, and CutMix lowers it sharply. Per class, CutMix makes Organic collapse ($0.447 \rightarrow 0.055$) along with Plastic, while making the Clothing attractor even stronger ($0.855 \rightarrow 0.890$): CutMix teaches mixed, low-confidence predictions, and out-of-distribution that uncertainty resolves towards the dominant class — the failure pattern of Section 3, amplified.

Conclusion. Taken at face value, this completes a clear-cut picture: photometric augmentation, rebalancing and now geometry as well all fail on RealWaste. However, we deliberately stop just short of this conclusion, for a precise reason: so far every judgment has been based on a single yardstick, RealWaste, chosen specifically because it is a pessimistic case. The assignment warns us that there will be different cameras and conditions, but does not tell us *how strong* this change will be; if we relied only on the severity of RealWaste, we would risk discarding a countermeasure that, under a lighter shift,

would instead have worked well. This is why in Section 8 we build an evaluation yardstick calibrated precisely on the assignment.

8 Recipe Selection via a Calibrated Internal Stress Test

Motivation. Clean validation fails to separate the recipes (all within ± 0.005 , because it is saturated); RealWaste may exaggerate the difficulty. We need a third, intermediate yardstick: the same images and labels as our validation, but presented under harder, assignment-like conditions, at more than one severity level.

Method. We build a *hard-validation set* by applying moderate, realistic corruptions to the validation images: framing and scale changes (crop/zoom, small rotations), resolution degradation, JPEG recompression, mild photometric jitter and light sensor noise, at two intensities (**mild**, **moderate**). Two properties keep the comparison fair and within the rules: no model is re-trained — the saved checkpoints are evaluated as-is; and the training set is never touched — only the evaluation is stressed.

Recipe	clean-val	hard-val (mild)	hard-val (moderate)	RealWaste (severe)
baseline (CE)	0.967	0.937	0.828	0.455
light	0.968	0.932	0.828	0.459
weighted_loss	0.961	0.936	0.802	0.453
geo_rrc (light + RRC)	0.964	0.946	0.860	0.426
geo_rrc + CutMix	0.965	0.938	0.871	0.331

First reading. The ranking depends on severity: starting from zero shift, **geo_rrc** leads under mild/moderate, and the order reverses at RealWaste. Taken at face value, this reading would overturn Section 7 and crown **geo_rrc** as the best recipe. This is the conclusion we had drawn in our first draft. In the next lines we will show, on the contrary, that reality is quite different.

A hidden bias (confounder) in the probe. Before committing to the choice, we asked ourselves *why* **geo_rrc** was in the lead — and we discovered that the comparison was premature. The aggregate hard-validation score is dominated by framing and scale perturbations, which are exactly the transformation **geo_rrc** trains on (RandomResizedCrop): the probe rewards the recipe for being robust to its own augmentation, which is almost circular. The aggregate also underweights the acquisition axis (the one the assignment names first: “different cameras, very different conditions, post-processing”) and never isolates the two conditions the assignment explicitly mentions, background removal and resolution change.

Assignment-aligned decomposition. We rebuilt the probe as four perturbation families aligned one-to-one with the assignment — *geometric* (framing/scale), *acquisition* (blur, JPEG, jitter, noise), *background* (subject isolated on a neutral field, “background removed”) and *resolution* — evaluated separately, at the same intensities, with the same fair protocol. Before reading the numbers we fixed a selection rule: the primary axis is acquisition; a minimum robustness threshold requires that the chosen recipe not be last in any family; clean-val must stay within ± 0.005 of the best; ties are resolved on the average of the four families.

Recipe	clean-val	geometric	acquis.	background	resol.	4-fam. avg.
light	0.968	0.957	0.920	0.887	0.932	0.924
strong	0.954	0.946	0.917	0.878	0.943	0.921
baseline (CE)	0.967	0.956	0.913	0.886	0.939	0.924
weighted_loss	0.961	0.956	0.912	0.844	0.927	0.910
sampler	0.962	0.951	0.909	0.857	0.929	0.912
geo_rrc	0.957	0.957	0.893	0.828	0.927	0.901
geo_rrc + CutMix	0.965	0.961	0.871	0.930	0.918	0.920

Table 3: 8-class Bal.Acc per family, averaged over mild and moderate. Recipes ranked by the acquisition axis; the selected recipe (light) is highlighted.

Reading. The decomposition overturns the first reading. The advantage of **geo_rrc** was confined

to its own “home” family — where it doesn’t even beat **light** (0.957 vs 0.957) — while on acquisition it is among the worst (0.893) and it is last on the average across all families and lowest on background, failing the minimum robustness threshold. **light** is first on acquisition (0.920) and first on the average across all families (0.924), never worse than any competitor on any axis, with the best clean-val; the independent RealWaste probe confirms it ($\text{light } 0.459 > \text{geo_rrc } 0.426$).

Decision. We adopt **light**. We reject **geo_rrc** (its advantage was an artifact of the biased probe), CutMix (last on acquisition, 0.331 on RealWaste) and **weighted_loss** (kept only as the diagnostic of Section 6). Even **light**, however, stops at 0.920 on acquisition — its weakest family and the primary axis of the assignment — leaving a clear margin that Section 10 will go on to fill. Under moderate acquisition the weakest classes are Metal (~ 0.75) and Plastic (~ 0.86): they become the targets of the backbone comparison. *Methodological lesson: a robustness test is reliable only if what it perturbs is independent of what the model was trained on.*

9 Architecture Comparison

9.1 Two axes, resolved sequentially

Recipe and architecture are independent axes, and exploring their full combination is neither necessary nor sustainable time-wise. Having fixed the recipe (**light**), we now fix the architecture, training each candidate with the identical recipe: any difference will therefore be attributable to the architecture alone.

9.2 What “best model” means here

Clean accuracy is saturated (0.96–0.98 for any reasonable choice) and carries almost no signal; the accuracy still in play is the one under shift. Moreover, the task explicitly evaluates a three-way trade-off: Balanced Accuracy, GPU memory at test time (less is better) and frame rate (more is better). Therefore: select primarily on hard-validation accuracy, treat clean validation as a sanity check, and prefer the best point on the frontier — a heavier model is justified only if it converts its cost into robust accuracy, especially on Metal and Plastic.

9.3 Candidates and exclusions

Four torchvision backbones pre-trained on ImageNet, chosen to cover the capacity spectrum: ResNet18 (11.7M parameters — the fast, lightweight incumbent), RegNetY-1.6GF (excellent accuracy/real-GPU-latency ratio), EfficientNetV2-S (higher capacity, designed to exploit the GPU well) and ConvNeXt-Tiny (28M, modern design, native at 224×224). Excluded, with their respective motivations: MobileNetV3 — it does not exploit the GPU well, so despite being a lightweight model it is slow at processing images (often slower than ResNet18, contrary to what the project requires); EfficientNetV2-M — it is too large and complex for the few images available, with a risk of overfitting, and it is designed for images too large for our system to handle.

9.4 Fair comparison protocol

Everything except the backbone remains identical to the frozen **light** configuration: same `split.csv`, 224×224 , same augmentation, plain Cross-Entropy, 15 epochs, cosine schedule, seed 1234, best-epoch checkpoint, same deterministic validation transform, full fine-tuning. Optimizer: SGD for three backbones, AdamW for ConvNeXt-Tiny, which is notoriously hard to fine-tune with plain SGD — we compare the architectures at their best rather than penalizing one for the wrong reason; we flag the absence of an in-depth per-backbone learning rate analysis as a limitation, since, having already achieved convergence on ResNet, we did not verify it on the other architectures. Background noise: repeated runs of the exact same configuration differ by ~ 0.007 (GPU non-determinism), consistent with the CV noise of ± 0.005 ; the decision rule must respect it.

9.5 The 5 GB training memory budget

At a physical batch of 64, two backbones exceed the 5 GB limit in training and RegNetY meets it only barely. Reducing the batch only for the heavy models would change the optimization dynamics

and compromise the fairness of the comparison. We solve both problems with *gradient accumulation*: the effective batch remains 64 for everyone, and only the physical micro-batch is adapted — the optimization recipe is identical, only the speed changes. Following this modification, all runs were verified to be under 5 GB by peak memory readings.

Backbone	VRAM @ batch 64	Within 5 GB?	Micro-batch $\times$ accum.
ResNet18	~ 1.94 GB	yes	64×1
RegNetY-1.6GF	~ 4.92 GB	borderline	32×2
ConvNeXt-Tiny	~ 7.43 GB	no	32×2
EfficientNetV2-S	~ 10.64 GB	no	16×4

9.6 Metrics and pre-registered decision rule

For each backbone we measure: clean-val Bal.Acc (sanity check only); hard-val Bal.Acc (mild + moderate, same assignment-aligned pipeline for everyone — the primary criterion); GPU memory at test time (peak during a forward pass; must stay under 4 GB); frame rate. Decision rule, fixed before reading the results: the backbone with the highest hard-val average wins; if the performance difference is minimal, with a gap of $\Delta \leq 0.01$ (about twice the normal run-to-run variation), we treat it as a tie and the more efficient model wins (prioritizing frame speed first and then memory footprint); in the case of a further tie, the choice falls on the simplest starting model (ResNet18).

9.7 Sanity check (clean validation)

Backbone	clean-val Bal.Acc	Metal TPR	Plastic TPR	Weights
RegNetY-1.6GF	0.985	0.974	0.965	~ 42 MB
EfficientNetV2-S	0.984	0.961	0.954	~ 82 MB
ConvNeXt-Tiny	0.977	0.987	0.954	~ 111 MB
ResNet18	0.972	0.942	0.919	~ 45 MB

All four trainings are valid. From the table just reported, however, two observations emerge: the larger models outperform ResNet18 by a smaller margin than expected (since the unperturbed dataset flattens the differences and does not, by itself, justify the cost of a heavier model), and the extra computational capacity visibly benefits the harder classes (RegNetY raises Plastic from 0.919 to 0.965; ConvNeXt brings Metal to 0.987) — an encouraging signal, but its actual resilience in the face of altered data (under shift) will be verified in the next step.

9.8 Robustness comparison and selection

Backbone	Hard-val (avg.)	Test mem. (GB)	Frame rate (fps)	Metal TPR (shift)	Plastic TPR (shift)
EfficientNetV2-S	0.964	0.47	280	0.831	0.861
RegNetY-1.6GF	0.963	0.30	356	0.831	0.827
ConvNeXt-Tiny	0.947	0.53	200	0.870	0.832
ResNet18	0.934	0.29	1186	0.779	0.856

The capacity gain survives — and grows — under shift. All three larger backbones beat ResNet18 by more than the background noise, and by more than they do on the clean set (RegNetY: +0.014 on clean, +0.029 on hard): the extra capacity converts into robust accuracy, the only justification allowed by Section 9.2.

Applying the decision rule: EfficientNetV2-S (0.9637) against RegNetY (0.9625) gives $\Delta = 0.0012$ — a statistical tie — so the decision passes to efficiency, where RegNetY dominates on both axes (356 vs 280 img/s; 0.30 vs 0.47 GB) with about half the parameters. Therefore, at equal accuracy, RegNetY is more efficient than EfficientNetV2-S: **RegNetY-1.6GF is thus selected as the winning backbone.** The ranking of the top two is identical if one relies solely on the acquisition axis instead of the four-family average, so the choice does not depend on the criterion. All candidates are well under

the 4 GB test budget: memory is not the binding axis; robustness and throughput are.

Honest limitation: RegNetY clearly improves Metal in the hardest condition ($0.779 \rightarrow 0.831$), but Plastic in that same condition (0.827) remains marginally below ResNet18 (0.856) despite a large advantage on clean — Plastic under hard acquisition remains the residual weak spot. ConvNeXt-Tiny is dominated (less robust, heavier, slower); ResNet18 remains the fallback for pure speed (1186 img/s), but its robustness deficit exceeds the background noise.

9.9 Test-Time Augmentation (TTA)

As a refinement applied only in the final stage, each image is evaluated together with its horizontally mirrored version and the two results are averaged. On our most severe test this raises the score from 0.9625 to 0.9645: a minimal change that falls within normal statistical noise, so we do not declare it a genuine improvement. We nonetheless keep it in the final code because it carries no risk: mirroring the image does not change the object (a mirrored bottle is still a bottle), so it cannot ruin predictions that are already correct; the only price to pay is speed (which drops from $\sim 356 \rightarrow \sim 178$ img/s, still comfortably within the required limits, while memory stays fixed at 0.30 GB). It is an extra safety margin, not a real performance gain.

10 Strengthening the Primary Axis: the `acq_mild` Recipe

Motivation. `light` stops at 0.920 on acquisition — the primary axis of the assignment and its own weakest family. Section 5 had identified the harmful end of the intensity curve (**strong**) but had never tested its central part: a mild, calibrated dose of acquisition-type perturbation, strong enough to break the capture shortcuts of Section 3, but light enough to preserve the color and glossiness cues. We hypothesize that this dose sits near the peak of the inverted U.

Design. `acq_mild` = `light` + a mild acquisition block (light Gaussian blur, mild simulated JPEG recompression, small photometric jitter, light sensor noise), applied only to the training set; everything else is frozen as in every recipe experiment. We stay on the ResNet18 test bench for comparability; the chosen backbone is re-trained under `acq_mild` in Section 12.

Recipe	clean-val	geom.	acquis.	background	resol.	4-fam. avg.	RealWaste
light	0.968	0.957	0.920	0.887	0.932	0.924	0.459
<code>acq_mild</code>	0.965	0.961	0.956	0.867	0.955	0.935	0.492

Result. `acq_mild` raises the primary axis from 0.920 to **0.956**, leads the average across all families, is best on resolution and ties the best on geometry, at no in-distribution cost. On the independent RealWaste probe it reaches **0.492** — the best of the entire study, +0.033 over `light` (~ 6 times the noise). The per-class pattern is exactly what a good recipe should produce: the Clothing attractor weakens ($0.824 \rightarrow 0.777$) and most minority classes gain (organic $0.336 \rightarrow 0.482$, glass, plastic, papery), with Metal as the only regression. Crucially, this is a net gain, not a self-cancelling redistribution as in **strong** or `weighted_loss`.

Self-check for the Section 8 bias. For methodological rigor we must welcome this result with caution: `acq_mild` is trained on acquisition disturbances and evaluated, in part, on the same category — a quasi-circularity similar to the one that had overestimated `geo_rrc`. Applying the same test to eliminate this bias (excluding the “home” category), the internal check shows a tie (0.928 vs 0.926). The final choice therefore rests on two pieces of evidence that this bias cannot explain: the advantage in the resolution category (on which `acq_mild` did no training) and, above all, the +0.033 gain obtained on RealWaste, a completely independent dataset.

Reconciliation with Experiment 1: no contradiction — Section 5 had found the descending stretch of the inverted U (**strong** overdoes it), this section finds its peak: the sign of the effect depends on the dose.

A more precise description of the gap. The gap has a recoverable acquisition component, which the calibrated augmentation reclaims (RealWaste $0.455 \rightarrow 0.492$), but it is not closed ($0.492 \ll 0.965$): the dominant residual is the object/context shift between clean product photos and real-world waste, beyond the reach of any data-side lever we found.

10.1 Two extensions that fail

acq_mild + RandomErasing. Hypothesis: an occlusion augmentation should help the “multiple objects / partial views” condition. Result: no gain on the geometric or background families, and RealWaste drops $0.492 \rightarrow 0.444$ (~ 9 times the noise). Mechanism: on a training set dominated by Clothing, erasing patches pushes the model to “fill in” towards the majority prior — the attractor re-inflates ($0.777 \rightarrow 0.836$) while the weakest minority classes get crushed (organic $0.482 \rightarrow 0.306$, plastic $0.465 \rightarrow 0.330$). *Rejected.*

acq_mild + class-weighted loss. Hypothesis: now that acq_mild has improved the representation, re-weighting might finally take effect. Result: acquisition drops ($0.956 \rightarrow 0.948$), background collapses ($0.867 \rightarrow 0.829$), RealWaste decreases $0.492 \rightarrow 0.476$. Re-weighting again trades TPR between classes with a net loss — confirming, at a finer grain, that Plastic and Metal are weak under shift for appearance reasons, not frequency ones. *Rejected.*

What these negative results establish together. The only intervention with a real, non-redistributive OOD gain is the one that improves *what the features encode* (acq_mild). Every intervention on the decision boundary, on the sampling prior, or on pasting/erasing patches — weighted loss, sampler, CutMix, RandomErasing — redistributes TPR with a net of zero or negative, or re-inflates the majority attractor. **Decision: acq_mild is frozen as the final recipe.**

11 Robustness to Multiplicity of Same-Class Objects

The assignment lists photos with multiple objects of the same class. We verify rather than train for this, for two reasons: our backbones end with a global average pooling, so multiple instances of a class reinforce a single aggregated signal instead of competing with each other; and multiplicity is already present in the training data (batteries in pairs, glass fragments, crates of bananas, stacks of magazines). Protocol (evaluation only, no re-training): from the validation images we compose 2×2 and 3×3 same-class mosaics on a white field and evaluate the winning recipe (ResNet18 acq_mild) against the single-object reference.

	single	2×2	3×3
Balanced accuracy	0.965	0.925	0.694
clothing	0.997	0.775	0.150
metal	0.935	0.875	0.375
plastic	0.902	0.825	0.475
battery / glass / papery / undiff.	≥ 0.97	≥ 1.00	≥ 0.93

Reading. Under mild multiplicity (2×2) the model is essentially invariant, and the natively multiple classes are perfect. The drop at 3×3 conflates two effects: multiplicity itself and a severe scale artifact — nine objects side by side in a 224 px frame shrink each instance several-fold. Indeed, the collapse concentrates exactly in the classes that require fine local detail (clothing 0.15, metal, plastic), while the classes recognizable by texture stay high (papery 1.00, glass 0.975). The realistic condition — a few same-class objects filling the frame — corresponds to the robust 2×2 regime; the limitation related to small objects/low resolution is recorded as a known weak spot consistent with the resolution family, not as a multiplicity failure. Thanks to the shared global-average-pooling design, we expect the same invariance on the delivered backbone.

12 Final Model and Conclusions

The delivered system is a **RegNetY-1.6GF fine-tuned under acq_mild**, chosen in three pre-registered stages — first the recipe family (light, Section 8), then the architecture (RegNetY, Section 9), finally the recipe strengthening (acq_mild, Section 10) — each judged on the de-confounded, assignment-aligned probe, not on the saturated clean score. Transferred to the chosen backbone, acq_mild reaches a hard-validation Balanced Accuracy of **0.971** (clean 0.980) and RealWaste **0.582**, with 0.30 GB of test memory and ~ 178 img/s — confirming that the gain on the acquisition axis transfers to the architecture, as predicted by the recipe/architecture separability of Section 4. The `predict()` of the test script loads `regnety16gf_acq_mild.pth`, applies the same deterministic `Resize(224) + Im-`

ageNet normalization used throughout the study, and averages the prediction over the input and its horizontal flip (TTA, Section 9.9).

12.1 Fine-tuning depth: partial freezing

One motivated lever had remained untested: training fewer layers, to protect the pre-trained features from distortion. We test it directly on the delivered model, changing a single factor (which layers get updated): `full_ft` (delivered, all layers), `shallow_freeze` (stem + blocks 1–2 frozen, $\sim 4\%$ of parameters), `mid_freeze` (block 3 as well, $\sim 57\%$), `lp_ft` (3 epochs head-only, then full fine-tuning).

Variant	clean-val	acquisition	hard-val avg.	RealWaste (OOD)
<code>full_ft</code> (delivered)	0.980	0.973	0.971	0.582
<code>shallow_freeze</code>	0.982	0.973	0.972	0.604
<code>mid_freeze</code>	0.971	0.948	0.950	—
<code>lp_ft</code>	0.975	0.968	0.964	—

Table 4: `mid_freeze` and `lp_ft` turned out to be dominated on the in-distribution probe and were not taken to the OOD probe.

Reading and decision. On the selection axis, `shallow_freeze` and `full_ft` tie ($\Delta = 0.0014 \ll 0.01$) and are identical on both efficiency axes. `mid_freeze` clearly regresses — block 3 must adapt to the waste domain — and `lp_ft` brings no benefit: the two bracket the useful freezing depth from both sides. The only advantage of `shallow_freeze` is on RealWaste (0.604 vs 0.582, above the noise), consistent with the preservation of generic low-level ImageNet features. Our pre-registered rule resolves the tie in favor of the incumbent, `full_ft`: RealWaste has been a confirmation probe throughout the study, never a selection criterion, and promoting it now would break the very discipline that makes the previous decisions reliable. `shallow_freeze` is recorded as the strongest candidate for a future iteration — a robustness gain at zero inference cost.

12.2 A post-hoc check: confidence-based reject option

Since Undifferentiated is the class most exposed to shift, we tested a refinement that requires no re-training: redirecting a prediction to Undifferentiated whenever the maximum softmax probability (after TTA) falls below a threshold τ . Two thresholds were pre-registered ($\tau = 0.5$, $\tau = 0.7$) instead of being swept, because sweeping τ on the same probe used to report performance would itself be tuning on the evaluation set (overfitting on the validation set).

Variant	hard-val avg.	acquisition	Metal TPR	Plastic TPR	Undiff. TPR
no reject	0.9724	0.9746	0.9481	0.9364	0.9496
$\tau = 0.5$	0.9701	0.9726	0.9351	0.9191	0.9640
$\tau = 0.7$	0.9601	0.9614	0.8961	0.8844	0.9712

Both thresholds trade Metal/Plastic TPR for Undifferentiated TPR — the same zero-sum pattern on the decision boundary seen before. $\tau = 0.7$ clearly exceeds the pre-registered minimum threshold; $\tau = 0.5$ stays within the limit but gains too little ($+0.014$) to be real. *Rejected:* `predict()` is delivered without it — a further confirmation that decision-boundary interventions redistribute TPR instead of creating it.

Key takeaways

- **In-distribution validation was saturated and uninformative:** the competition is decided on robust accuracy under distribution shift, which we probed with an assignment-aligned internal stress test, used together with the deliberately pessimistic lens of RealWaste.
- **A single recipe-side lever produced a real, non-redistributive OOD gain:** the mild, calibrated acquisition augmentation (`acq_mild`), which takes RealWaste from 0.455 to 0.492 and the assignment’s primary axis from 0.920 to 0.956, weakening the Clothing attractor. The effect is dose-dependent: aggressive corruption overdoes it and hurts; the calibrated dose helps.

- **Every intervention on the decision boundary or the prior failed** — loss re-weighting (alone and on top of `acq_mild`), oversampling, CutMix, RandomErasing and the reject option — redistributing TPR with a net of zero or negative, or re-inflating the majority attractor. The lever is what the features encode, not where the boundary sits; the dominant residual is the object/context shift, which `acq_mild` reaches only in part.
- **Our most instructive methodological result is a hidden evaluation error we discovered in our own testing system:** a global stress evaluation ended up rewarding the method trained precisely on that type of transformation (`geo_rrc`). Splitting the test into separate categories according to the project’s requirements, following a rule set in advance, overturned the final choice — a robustness test is reliable only if there is real independence between the disturbances used to evaluate and the data the model was trained on.
- **Greater capacity converts into robust, not just clean, accuracy:** RegNetY’s advantage over ResNet18 widens under shift, and adds to that of the recipe (RealWaste 0.582 vs 0.492 for ResNet18 under `acq_mild`). Plastic under the hardest acquisition remains the residual weak spot.